Project Overview

Using data from **400 videos across 10 performance metrics**, this project analyzes how different factors influence short-form video success. It examines patterns in hook strength, niche, early engagement, retention, total views, music type, and upload timing to uncover what drives stronger audience performance and helps guide smarter content strategies.

Dataset Summary

- **Data Size:** 400 rows × 10 columns
- **Includes:**
- Video metadata (video_id, duration_sec, niche, upload time)
- Performance metrics (view_first_hour, views_total, retention_rate)
- Engagement features (hook_strength_score, first 3 sec engagement)
- Content attributes (music type)

Python-Based Exploratory Data Analysis

Workflow Overview – Data Preparation and Cleaning in Python

We began the workflow by preparing and exploring the dataset in Python:

- **Data Loading:** The dataset was imported using *pandas* for further processing.

- **Initial Exploration:** Ran `df.info()` along with basic summary checks to understand the dataset's structure, column types, and key statistics.

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 10 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   video_id               400 non-null    object 
 1   duration_sec           400 non-null    int64  
 2   hook_strength_score    400 non-null    float64
 3   niche                  400 non-null    object 
 4   views_first_hour       400 non-null    int64  
 5   views_total            400 non-null    int64  
 6   retention_rate         400 non-null    float64
 7   first_3_sec_engagement 400 non-null    float64
 8   music_type             400 non-null    object 
 9   upload_time            400 non-null    object 
dtypes: float64(3), int64(3), object(4)
memory usage: 31.4+ KB
```

● **Data Quality Check:** Verified that the dataset contained **no missing values**, ensuring it was ready for analysis without additional cleaning steps.

```python
# checking for null values

df.isnull().sum()
```

```
video_id                  0
duration_sec              0
hook_strength_score       0
niche                     0
views_first_hour          0
views_total               0
retention_rate            0
first_3_sec_engagement    0
music_type                0
upload_time               0
dtype: int64
```

- Created a new column to categorize video lengths into **Short**, **Medium**, and **Long** based on their duration in seconds, enabling clearer comparisons across content types.

```
# Made a new Column Duration categories as short medium and long to categorize video durations

df['duration_categories']=pd.qcut(df['duration_sec'],q=3,labels=['short','medium','long'])
df[['duration_categories','duration_sec']]
```

	duration_categories	duration_sec
0	short	16
1	short	19
2	short	13
3	long	36
4	short	8

- made a growth percentage column to calculate growth percentage, some rows exceeded the bracket of 100 percent clearly it had a data entry fault.

```
#made a growth percentage column to calculate growth percentage

df['growth_percentage']=df['views_first_hour']/df['views_total']*100
df.head()
```

hook_strength_score	niche	views_first_hour	views_total	retention_rate	first_3_sec_engagement	music_type	upload_time	duration_categories	growth_percentage
0.61	Motivation	10695	886048	0.85	0.26	Viral Track	2024-02-09	short	1.207045
0.34	Tech	1812	198243	0.56	0.80	Trending	2024-04-17	short	0.914030
0.47	Travel	23247	584171	0.14	0.27	Trending	2024-01-01	short	3.979485
0.21	Travel	11515	491291	0.29	0.81	Remix	2024-04-01	long	2.343825
0.75	Gaming	13474	607425	0.27	0.44	Viral Track	2024-01-15	short	2.218216

[25]:
```
# Some rows exceeded the bracket of 100 percent clearly it had a data entry fault

df[df['growth_percentage']>100].sort_values(by='growth_percentage',ascending=False)
```

[25]:

sec	hook_strength_score	niche	views_first_hour	views_total	retention_rate	first_3_sec_engagement	music_type	upload_time	duration_categories	growth_percentage
8	0.99	Fitness	17084	1924	0.83	0.80	No Music	2024-10-05	short	887.941788
19	0.94	Music	31891	5716	0.45	0.60	Original	2024-11-17	short	557.925122
25	0.75	Fitness	23725	5901	0.62	0.32	Viral Track	2024-01-18	medium	402.050500
8	0.72	Tech	39685	10420	0.85	0.73	Trending	2024-02-10	short	380.854127
43	0.55	Beauty	23378	8162	0.71	0.32	Viral Track	2024-06-28	long	286.424896
29	0.43	Tech	26423	10666	0.25	0.30	Original	2024-01-02	medium	247.731108
21	0.42	Travel	27246	12678	0.49	0.51	Viral Track	2024-07-11	medium	214.907714

- Some rows exceeded the bracket of 100 percent clearly it had a data entry fault, swapped values which exceeded higher percentage considering it as a data entry mistake and recalculated growth percentage

```
# swapped values which exceeded higher percentage considering it as a data entry mistake
a=df['growth_percentage'] > 100
df.loc[a, ['views_first_hour', 'views_total']] = (df.loc[a, ['views_total', 'views_first_hour']].values)
```

```
# recalculating growth percentage
df['growth_percentage']=df['views_first_hour']/df['views_total']*100
```

```
[33]: df[['views_total', 'views_first_hour','growth_percentage']].reset_index().sort_values(by='growth_percentage',ascending=False)
```

[33]:

	index	views_total	views_first_hour	growth_percentage
309	309	37944	35895	94.599937
36	36	6860	6483	94.504373
278	278	25462	22034	86.536800
112	112	40037	31761	79.329121
356	356	35583	27655	77.719698

- made a new column for better categorization of growth , and growth description.

```
[37]: df[['growth_percentage','growth_category']].head()
```

[37]:

	growth_percentage	growth_category
0	1.207045	late_viral
1	0.914030	late_viral
2	3.979485	balanced_growth
3	2.343825	late_viral
4	2.218216	late_viral

```
[39]: df['growth_category_description']=df['growth_category'].map(
    {'late_viral':'Not strong initially goes viral later',
     'balanced_growth':'performs consistently over time',
     'early_growth':'good start moderate traction',
    'Flash_viral':'strong first hour then gradually declines'}
)
```

```
[41]: df[['growth_percentage','growth_category','growth_category_description']].sort_values(by='growth_percentage',ascending=False).head()
```

[41]:

	growth_percentage	growth_category	growth_category_description
309	94.599937	Flash_viral	strong first hour then gradually declines
36	94.504373	Flash_viral	strong first hour then gradually declines
278	86.536800	Flash_viral	strong first hour then gradually declines
112	79.329121	Flash_viral	strong first hour then gradually declines
356	77.719698	Flash_viral	strong first hour then gradually declines

Connecting to MYSql for deeper analysis

- Loaded the cleaned dataset into MySQL using a Python–database connection.

```python
from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "admin"
host = "localhost"
port = "3306"
database = "viral_reels"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

# Write DataFrame to MySQL
table_name = "reels_analysis"
df.to_sql(table_name, engine, if_exists="replace", index=False)
```

SQL Analysis

- Niches generating highest revenue potential per video based on weighted factors

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

niche	total_videos	Avg_views	Avg_hook_strength	Avg_retention_rate	Revenue_rate
Motivation	30	571626.33	0.59	0.52	342976.03
Food	50	519023.64	0.61	0.54	311414.42
Education	40	503269.50	0.62	0.55	301961.94
Comedy	39	496791.46	0.57	0.56	298075.1
Fitness	41	485828.32	0.6	0.49	291497.22

- Video durations that produces the most late viral video (views scales after an hour)

duration_categories	Avg_duration	Total_videos	count_of_late_viral	Pct_of_late_viral
short	11.52	137	40	29.20
medium	26.04	134	36	26.87
long	38.50	129	24	18.60

- High retention videos with below average first hour views (missed monetization opportunities)

video_id	duration_sec	hook_strength_	niche	views_first_hour	views_total	retention_rate	first_3_sec_	music_type	upload_time	duration_categories	growth_percentage	growth_category	growth_category_description
vid_38	10	0.25	Beauty	11852	131978	0.94	0.61	Viral Track	2024-10-04	short	8.98028459288...	early_growth	good start moderate traction
vid_75	33	0.75	Gaming	16129	420673	0.94	0.84	No Music	2024-05-21	long	3.83409441537...	balanced_growth	performs consistently over time
vid_23	43	0.76	Music	2902	80523	0.93	0.78	Original	2024-08-24	long	3.60393924717...	balanced_growth	performs consistently over time
vid_393	30	0.27	Travel	12600	479812	0.93	0.75	Remix	2024-10-10	medium	2.62602852784...	balanced_growth	performs consistently over time
vid_127	28	0.91	Beauty	21548	802106	0.92	0.72	Remix	2024-09-28	medium	2.68642797834...	balanced_growth	performs consistently over time

- Music type generating the highest revenue potential (late viral growth)

music_type	total_videos	Avg_growth_PCT	late_viral_videos
Original	87	8.02	23
No Music	78	8.32	22
Trending	86	8.55	19
Viral Track	73	13.21	18
Remix	76	12.01	18

- Upload day with the highest ROI per video patterns

Upload_day	ROI
Wednesday	310975.4
Sunday	302984.36
Monday	301134.82
Tuesday	290658.63
Thursday	285419.61
Saturday	282203.17
Friday	273357.84

- Detecting underperforming niches (below average views)

niche	AVG_views
Tech	428601.33
Travel	473279.23
Gaming	484817.98
Music	461631.76
Fitness	485828.32
Beauty	465505.23

- Finding niches that convert high hook strength into actual retention

niche	hook_strength	Retention
Beauty	0.58	0.6

- Top 3 music type for every niche

	niche	music_type	views_total	Ranking
▶	Beauty	Remix	988262	1
	Beauty	Original	929848	2
	Beauty	No Music	843149	3
	Comedy	Remix	983717	1
	Comedy	Trending	936614	2
	Comedy	No Music	912332	3
	Education	No Music	990186	1
	Education	Original	950670	2
	Education	Remix	930345	3
	Fitness	Viral Track	983266	1
	Fitness	No Music	968547	2
	Fitness	Original	959996	3
	Food	Remix	981905	1
	Food	Remix	977443	2
	Food	No Music	950499	3
	Gaming	No Music	992963	1
	Gaming	No Music	992435	2
	Gaming	No Music	988260	3
	Motivation	Viral Track	979910	1
	Motivation	Original	969927	2

Power Bi Dashboard (for Visual Analysis)